

Multi Echelon Optimization- Plan of Action

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Week 1: Supply Chain and Demand Basics

Review core supply chain concepts such as echelons, warehouses, retailers, inventory flow, holding cost, stockout cost, ordering cost, and transportation cost. Simulate basic demand patterns such as normal and Poisson demand using Python.

Deliverable:

Demand simulation notebook

Week 2: EOQ and Linear Programming

Study the Economic Order Quantity model and understand its role in basic inventory planning. Learn the basics of Linear Programming and solve small LP problems using PuLP.

Deliverable:

EOQ solver and basic LP problem set

Week 3: Two-Echelon Supply Chain Model

Build a simple two-echelon model involving one warehouse and one retailer. Formulate inventory balance constraints, demand satisfaction constraints, ordering decisions, and cost minimization as a Linear Programming problem.

Deliverable:

Two-echelon LP model

Week 4: Multi-Period Planning with Lead Times

Extend the two-echelon model to multiple time periods. Add time-indexed inventory variables and incorporate lead times to make the replenishment planning more realistic.

Deliverable:

Multi-period model with lead times

Week 5: Multiple Retailers and Capacity Constraints

Expand the model by adding multiple retailers. Include supply capacity, warehouse capacity, and retailer-level constraints. Study shadow prices and interpret basic dual variables to understand the effect of capacity limitations.

Deliverable:

Multi-retailer LP model with shadow price analysis

Week 6: Stochastic Optimization and Midterm Checkpoint

Introduce uncertainty through scenario-based stochastic optimization. Create different demand scenarios and compare expected costs across them. Analyze how uncertainty affects inventory and ordering decisions.

Midterm Checkpoint: 3rd week of June**Deliverable:**

Stochastic model and cost comparison report

Week 7: Robust Optimization

Explore robust optimization formulations for uncertain demand. Compare deterministic, stochastic, and robust approaches, and analyze the trade-off between lower cost and better protection against uncertainty.

Deliverable:

Robust model and trade-off analysis

Week 8: Final Experiments and Report

Run final experiments, including possible disruption cases such as reduced supply, delayed replenishment, or limited warehouse capacity. Compile all models, results, observations, and learnings into the final report.

Deliverable of Week 8:

Final report & code submission

Resources:

[Linear Programming with Python and PuLP – Ben Alex Keen](#)

[Optimization with PuLP — PuLP 4.0.0a6 documentation](#)

[Management - Operations and Supply Chain Management - YouTube](#)